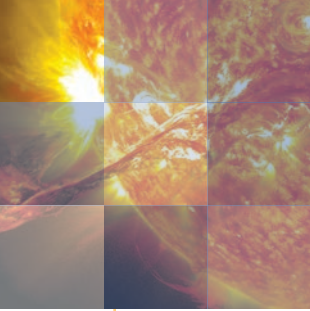




Reciprocal Power Functions

A power function is a function of the form $f(x) = ax^n$, where $a \neq 0$ and n is a positive integer.

The functions $f(x) = x$, $g(x) = x^2$, and $h(x) = x^3$ are power functions.

Graphing a Reciprocal Power Function

Definition

A **reciprocal power function** is a power function that has the power of x in the denominator of a rational function. The functions $f(x) = \frac{1}{x}$, $g(x) = \frac{1}{x^2}$, and $h(x) = \frac{1}{x^3}$ are reciprocal power functions.

EXAMPLE 1

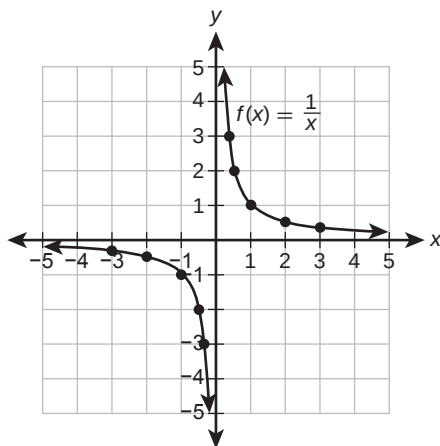
Graph $f(x) = \frac{1}{x}$.

SOLUTION

Use a table to find several ordered pairs in the function. Include some fractional values between $x = -1$ and $x = 1$ so that you know what happens on either side of and near $x = 0$, where $f(x)$ is undefined.

x	-3	-2	-1	$-\frac{1}{2}$	$-\frac{1}{3}$	0	$\frac{1}{3}$	$\frac{1}{2}$	1	2	3
$f(x) = \frac{1}{x}$	$-\frac{1}{3}$	$-\frac{1}{2}$	-1	-2	-3	undefined	3	2	1	$\frac{1}{2}$	$\frac{1}{3}$

This graph is called a rectangular hyperbola. The two distinct sections of the graph are called branches.



The function $f(x) = \frac{1}{x}$ is not defined for $x = 0$, so zero is excluded from the domain of the function. Notice that the graph gets closer and closer to the x -axis as the x -values approach positive infinity and negative infinity. Also, the graph gets closer and closer to the y -axis as the x -values get closer to zero. ■

► **Think About It** “ $f(x)$ approaches b ” means the values of $f(x)$ get closer and closer to b .

“ $f(x)$ approaches ∞ ” means the values of $f(x)$ increase without bound.

“ $f(x)$ approaches $-\infty$ ” means the values of $f(x)$ decrease without bound.

Definitions

The line $y = b$ is a **horizontal asymptote** of the graph of the function f if $f(x)$ approaches b as x approaches ∞ or $-\infty$.

The line $x = a$ is a **vertical asymptote** of the graph of the function f if $f(x)$ approaches ∞ or $-\infty$ as x approaches a , either from the left or the right.

Therefore, for $f(x) = \frac{1}{x}$, the x -axis (or $y = 0$) is a horizontal asymptote, and the y -axis (or $x = 0$) is a vertical asymptote.

Finding the Domain and Range of a Reciprocal Power Function

EXAMPLE 2

Determine the domain and range of the function.

A $f(x) = \frac{3}{x^2}$

SOLUTION

Find domain restrictions by setting the denominator equal to zero.

$$\begin{aligned}x^2 &= 0 \\x &= 0\end{aligned}$$

If $x = 0$, then the denominator equals 0. So the domain is the set of all real numbers except 0. In set notation, the domain is

$$\{x \mid x \in \mathbb{R} \text{ and } x \neq 0\}.$$

Use some reasoning about the function to determine the range. Since x is squared, $f(x)$ can never be negative and $f(x)$ cannot equal 0. So the range is the set of all real numbers greater than 0,

$$\{y \mid y > 0\}.$$

B $g(x) = \frac{5}{2x^3}$

SOLUTION

Find domain restrictions by setting the denominator equal to zero.

$$2x^3 = 0$$

$$x^3 = 0$$

$$x = 0$$

The domain is the set of all real numbers except 0. In set notation, the domain is

$$\{x \mid x \in \mathbb{R} \text{ and } x \neq 0\}.$$

Use some reasoning about the function to determine the range. Since x is cubed, $g(x)$ can be negative or positive and $g(x)$ cannot equal 0. So the range is the set of all real numbers except 0,

$$\{y \mid y \in \mathbb{R} \text{ and } y \neq 0\}. \blacksquare$$

All reciprocal power functions of the form $f(x) = \frac{a}{x^n}$, where n is even, have the same domain and range. The domain is $\{x \mid x \in \mathbb{R} \text{ and } x \neq 0\}$. When a is positive, the range is $\{y \mid y > 0\}$, and when a is negative, the range is $\{y \mid y < 0\}$.

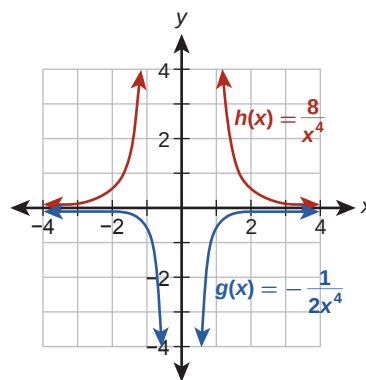
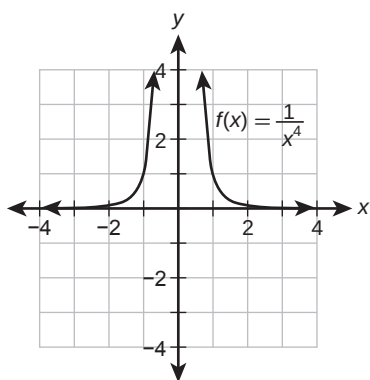
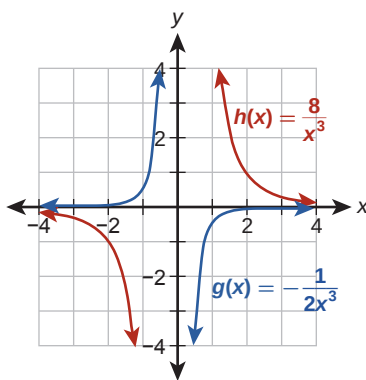
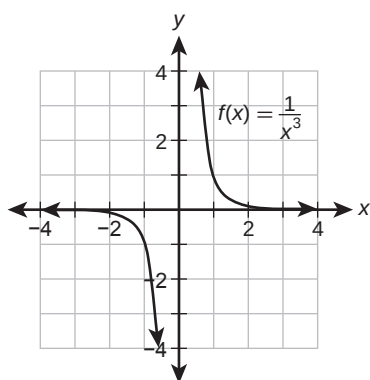
All reciprocal power functions of the form $f(x) = \frac{a}{x^n}$, where n is odd, have the same domain $\{x \mid x \in \mathbb{R} \text{ and } x \neq 0\}$ and range $\{y \mid y \in \mathbb{R} \text{ and } y \neq 0\}$.

Recognizing the Reciprocal Power Function Family of Graphs

Reciprocal Power Function Graph Family

Reciprocal Power Function Graph Family: $f(x) = \frac{a}{x^n}$

Equations of the form $f(x) = \frac{a}{x^n}$ are reciprocal power functions with a horizontal asymptote at $y = 0$ and a vertical asymptote at $x = 0$.



These graphs are graphs of the parent functions.

If you change the parameter a , you get curves that vary in their distance from the origin.

A negative a -value will reflect the graph across the x -axis.

Reciprocal power functions with odd powers have the same general shape, and reciprocal power functions with even powers have the same general shape. All reciprocal power functions have the x - and y -axes as asymptotes.

Graphing a Simple Reciprocal Power Function

EXAMPLE 3

Graph $f(x)$ and $g(x)$ on the same coordinate system. Compare the two graphs.

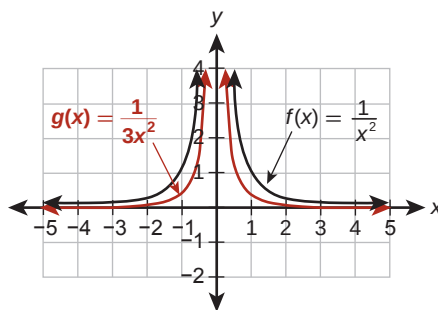
$$f(x) = \frac{1}{x^2}$$

$$g(x) = \frac{1}{3x^2}$$

SOLUTION

In both graphs, the domain is the set of all real numbers except 0.

$$\{x \mid x \in \mathbb{R} \text{ and } x \neq 0\}$$



- **Think About It** Values for x that make the denominator zero are excluded from the domain of $f(x)$. Since these are places where $f(x)$ is not defined, they may indicate a vertical asymptote. In functions such as $f(x) = \frac{1}{x^2}$, the value of $f(x)$ is always greater than zero, which may indicate a horizontal asymptote at $y = 0$.

Each graph has two branches, one for positive x -values and one for negative x -values. Each graph has a vertical asymptote whose equation is $x = 0$ and a horizontal asymptote whose equation is $y = 0$. ■

Asymptotes of Reciprocal Power Functions

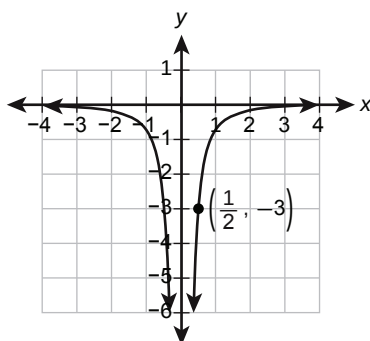
The graph of the function $f(x) = \frac{a}{x^n}$ has the following:

- vertical asymptote: y -axis ($x = 0$)
- horizontal asymptote: x -axis ($y = 0$)

Finding the Equation from a Graph

EXAMPLE 4

Find the equation for the reciprocal power function of the form $f(x) = \frac{a}{x^n}$ that has this graph.



SOLUTION

$$f(x) = \frac{a}{x^2}$$

Start with the general form of the equation.

$$-3 = \frac{a}{\left(\frac{1}{2}\right)^2}$$

Substitute the coordinates of the known point.

$$-\frac{3}{4} = a$$

Solve for a .

The equation of the function is $f(x) = -\frac{3}{4x^2}$. ■